

Written examination date: March 4th, 2023

Course title: Introduction to reinforcement learning and control

Course number: 02465

Aids allowed: All aids allowed

Exam duration: 4 hours

Weighting: The exam is divided into 3 parts:

- Multiple-Choice questions
- Conceptual questions
- Programming questions

The overall scores of each part contribute roughly equally towards the overall result. Each question in each part contribute equally towards the score of that part.

Part I: Questions 1-4 are multiple choice. The score of a correct answer is 3 points. The score of an incorrect answer is -1 points. The score of option E or blank is 0 points.

Part II and part III: Each completed sub-task contribute towards your score.

Preparing your handin: The three parts are prepared as follows:

Part I: Edit the file `irlc/exam/midterm2023a/multiple_choice_answers.py`. Don't include calculations. Only answer with `'A'`, `'B'`, `'C'`, `'D'`, `'E'`.

Part II: Create a PDF file with your answers and justifications.

Part III: Program your answer in the `.py`-files indicated in the question and run `irlc/exam/midterm2023a/midterm2023a_tests_grade.py` to generate your `.token`-file.

Handing in: To hand in, you should upload the files:

- The `irlc/exam/midterm2023a/multiple_choice_answers.py` -file with your answer to part I
- The `.pdf` -file with your answers to part II
- The `.token` -file with your answers to part III
- The `irlc/exam/midterm2023a/question_dp.py` source file containing your solution
- The `irlc/exam/midterm2023a/question_pid.py` source file containing your solution

Note on part II: The main quantities asked for are highlighted as $f(x)$. Answer unambiguously, concisely, and if applicable with algebraic simplifications. Your final result must be clearly indicated at the end of your answer: $f(x) = 3 \sin(x)$. To get credit, you must state the relevant theory and equations, and all relevant calculations must be included. Credit is not given for answers with missing or erroneous justifications.

Note on part III: To get started, move the folder `irlc/exam/midterm2023a` from the `.zip` file to the corresponding location in your existing exercise directory. The `.py` source files must be **reproducible** and **readable** so that someone else can run and fully understand your solution. You can freely use the `irlc`-toolbox (including solutions) and the packages we have used in the course, but you **must** include additional code you write during the exam or have prepared beforehand in the source files listed above. The source files must not be renamed. The `.token` file contains your results and must be up to date with your source files, i.e., generate it just prior to handin. In the case they differ, the `.token` file takes precedence. Credit is given for correct implementations defined by the problem description. The points in the `.token` file name is computed from the public tests, and might not correspond to overall correctness.

Part I: Multiple-Choice

Question 1:

Which one of the following options are correct?

- A. A PID controller is an example of an open-loop controller
- B. The integral parameter K_i must always be positive $K_i \geq 0$
- C. A PID controller is a model-based control method
- D. We can build a PID controller without specifying a cost-function**
- E. Don't know.

Solution: The correct answer is D.

- A. PID control is not open loop since it depends on the state
- B. The sign of the integral term (or any of the other terms) depends on the definition of u
- C. The PID controller does not require a model.
- D. A PID controller does not need a model or a cost function.**

Question 2:

Which one of the following options are correct?

- A. Control problems where the continuous-time dynamics takes the form $\ddot{x} = a\dot{x} + bx + c + u$ falls outside the scope of the linear quadratic regulator
- B. The linear-quadratic regulator is an example of model-free control
- C. In a linear-quadratic control problem of the form $x_{k+1} = Ax_k + Bu_k$, the matrices A and B must both be square.
- D. The cost-functions suitable for a linear-quadratic regulator can potentially produce negative values**
- E. Don't know.

Solution: The correct answer is D.

- A. The problem is linear if we use the state-representation $\mathbf{x} = \begin{bmatrix} x \\ \dot{x} \end{bmatrix}$
- B. The linear-quadratic regulator explicitly plans using a model of the dynamics and cost (hence the name, linear-quadratic)
- C. A is square but B is only square when x_k and u_k have the same dimension.
- D. Since the cost-function allows a constant term it can easily be negative.**

Question 3:

Which one of the following options are correct?

- A. Euler-discretization is necessary if we want to apply LQR to a general continuous-time control problem.
- B. Euler-discretization is exact when both the dynamics and cost-function for the continuous-time control problem are linear
- C. To apply Exponential integration to a control problem of the form $\dot{x} = Ax + Bu$, it is necessary that A is invertible

Question 3 continues on the next page...

- D. Euler-discretization can be applied to time-dependent control problems
- E. Don't know.

Solution: The correct answer is D.

- A. Finite-horizon LQR can be applied to any problem with linear dynamics $x_{k+1} = Ax_k + Bu_k$ without any reference to where these matrices come from (for instance, they can arise from exponential integration).
- B. This is not true since; c.f. the Harmonic Oscillator.
- C. A does not need to be invertible. In fact A is not invertible in the case $\ddot{x}(t) = u(t)$ (see exercises to lecture 4).
- D. Euler-discretization is formulated for time-dependent control problems. Recall that the update is just $x_{k+1} = x_k + \Delta f(x_k, u_k, t_k)$.

Question 4:

Suppose the Dynamical Programming algorithm is applied to a problem where the following is known:

- $N = 10$
- The size of the action spaces are $|\mathcal{A}_k(x_k)| = 4$
- The size of the states spaces are $|\mathcal{S}_0| = 1$, $|\mathcal{S}_N| = 2$ and otherwise $|\mathcal{S}_k| = 10$
- There are exactly two random noise disturbances, $w = 0$ and $w = 1$, available in any state/action combination:

$$P_W(w = 0|x, u) = P_W(w = 1|x_k, u_k) = \frac{1}{2}.$$

How many times does the dynamical programming algorithm need to evaluate f_k in order to find the optimal policy?

- A. 736
- B. 744
- C. 730
- D. 728**
- E. Don't know.

Solution: The correct answer is D.

At each time step $k = 0, \dots, N - 1$, the total number of evaluations of f_k is:

$$|\mathcal{S}_k| \times \text{actions} \times \text{disturbances}$$

The total is therefore $(1 + (N - 1) \times 10)(4 \times 2) = 728$

Part II: Conceptual questions

Question 5:

Consider a control problem where a control signal $u(t) \in \mathbb{R}$ is applied to a variable $w(t) \in \mathbb{R}$. The variable measures an angle, and it satisfies the following differential equation:

$$\ddot{w} = \cos(u + w) \quad (1)$$

We introduce a state $\mathbf{x}(t) = \begin{bmatrix} w(t) \\ \dot{w}(t) \end{bmatrix}$ which allows us to re-write the system in the usual way as a first-order differential equation:

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), u(t)).$$

The problem is then discretized using Euler discretization with a time step of $\Delta = 0.2$ to give states $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \dots$. Our goal is to bring the system to an angle $w = \frac{\pi}{2}$ (where it should remain), corresponding to the goal state $\mathbf{x}^* = \begin{bmatrix} \frac{\pi}{2} \\ 0 \end{bmatrix}$.

- (a) If we succeed at bringing the system to the target state at time t' , how much control $u(t')$ do we subsequently need to apply to keep the system at $w(t) = \frac{\pi}{2}$? Provide an argument for your answer (Hint: Consider what it would mean for the system to stand still in terms of $\mathbf{x}(t)$)

Solution: At the goal state, the derivative of $\mathbf{x}$ is $\begin{bmatrix} 0 \\ \cos(\frac{\pi}{2} + u(t)) \end{bmatrix}$. So assuming we don't apply any control the derivative is 0 and the system stays put. Therefore $u(t') = 0$

- (b) Assume that the initial conditions at the starting time $t_0 = 0$ is $w(t = 0) = \dot{w}(t = 0) = 0$ and that no control signal is applied to the system ($u(t) = 0, t \geq 0$). According to Euler discretization, what is the value of w at time $t_k = \Delta$?

Solution: Euler discretization is:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \Delta \begin{bmatrix} x_1 \\ \cos(x_1 + u_k) \end{bmatrix}.$$

This means that for $k = 0$ we get

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \Delta \begin{bmatrix} x_1 \\ \cos(0) \end{bmatrix} = \begin{bmatrix} 0 \\ \Delta \end{bmatrix}$$

so that $w_0 = 0$

- (c) Continuing the previous problem and still assuming no control signal is applied. According to Euler discretization, what is the value of w at time $t_k = 2\Delta$?

Solution: Continuing the above, at $k = 1$ we get:

$$\mathbf{x}_{k+1} = \begin{bmatrix} 0 \\ \Delta \end{bmatrix} + \Delta \begin{bmatrix} \Delta \\ \cos(0) \end{bmatrix} = \begin{bmatrix} \Delta^2 \\ 2\Delta \end{bmatrix}$$

I.e. $w_1 = \Delta^2 = 0.04$.

Question 6:

Consider the dynamical programming setting where we plan over a horizon $N > 0$. We consider a problem where:

- The terminal cost function is

$$g_N(x_N) = x_N^2$$

Question 6 continues on the next page...

- For all $k = 0, \dots, N - 1$ the dynamics is $f_k(x, u, w) = ax + u - \lambda w$
- The noise disturbances are normally distributed with variance σ^2 :

$$P_W(w|x, u) = \mathcal{N}(w \mid \mu = 0, \sigma^2 = 1)$$

- The states and actions are real numbers $\mathcal{S}_k = \mathcal{A}_k(x_k) = \mathbb{R}$.
- The non-terminal costs are only affected by u , $g_k(x, u, w) = u^2$.

Thus, the relevant parameters of the problem are a and λ . We are concerned with optimal control.

- (a) Although the problem has been formulated as being about dynamical programming, note that the structure of the problem is that of a 1-dimensional LQR model. In the case where $\lambda = 3$, derive the expected future cost $\mathbb{E}[g_N(x_N)|x_{N-1} = 0, u_{N-1} = 1]$ if we at time step $k = N - 1$ are in state $x_{N-1} = 0$ and take action $u = 1$

Solution: The future cost is given by

$$\mathbb{E}[x_N^2] = \mathbb{E}_{\mathcal{N}(w|0,1)} [(ax + u - \lambda w)^2] = \mathbb{E} [(1 + \lambda w)^2] = 1 + \lambda^2 \mathbb{E}[w^2] + \mathbb{E}[2\lambda w] = 1 + \lambda^2$$

so the answer is $\mathbb{E}[g_N(x_N)|x_{N-1} = 0, u_{N-1} = 1] = 10$

- (b) Derive an analytical expression for the optimal policy $\mu_k(x_k)$ in time step $k = N - 1$

Solution: Since it is known from the lecture notes that the optimal policy is not affected by noise we can set $\lambda = 0$. To proceed, we can either insert and optimize or run the LQR algorithm for 1 step. Using the LQR algorithm we get that:

$$V_N = Q_N = 2$$

$$R_k = 2 = R$$

$$A_k = a$$

$$B_k = 1$$

$$L_{N-1} = -(R_k + B_k^T V_{k+1} B_k)^{-1} (B_k^T V_{k+1} A_k) = -\frac{aQ_N}{R + Q_N} = -\frac{2a}{2 + 2} = -\frac{a}{2}$$

$$u_{N-1} = L_{N-1}x$$

Therefore $\mu_{N-1}(x) = -\frac{a}{2}x$

Part III: Programming questions

Question 7:

To solve this question, you should edit the file `irlc/exam/midterm2023a/question_dp.py`. This problem exclusively focuses on the inventory control problem discussed in [Her24, section 5.1.2]. The inventory control model and the dynamical programming algorithm is included in the exam folder.

The dynamical programming algorithm determines the optimal cost-to-go function J_k^* for a dynamical programming problem as a list of dictionaries. We are interested in building a variant of the DP algorithm which computes the expected tail cost of a policy $J_{\pi,k}(x_k)$, also represented as a list of dictionaries.

- The problem itself will be represented as a `DPModel` instance
 - The policy $\pi = (\mu_0, \mu_1, \dots, \mu_{N-1})$ is represented in the usual way as a list of length $N - 1$. Each element of this list corresponds to a μ_k and is represented as a dictionary which maps states to actions
- (a) Complete `def a_expected_items_next_day(x : int, u : int)`: This function is given the starting state x_0 and the first action u_0 , and computes the expected value of the next state x_1 :

$$\mathbb{E}[x_1 | x_0, u_0].$$

I.e., the function computes the expected amount of goods in the warehouse on day 1 given information about how much was in the warehouse on day 0 and how much we ordered u_0 . *Hint: Recall that $x_1 = f_0(x_0, u_0, w_0)$ where w_0 is the random noise disturbance at time step $k = 0$.*

- (b) Complete `def b_evaluate_policy(pi : list, x0 : int)`: This function is given a policy π in the before-mentioned format and a starting state x_0 , compute the expected tail cost $J_{\pi,0}(x_0)$ when starting in state x_0 at time $k = 0$ and subsequently taking actions according to π . *Hint: You may find [Her24, section 6.3.1] to be of help.*

Solution:

The solution can be found in this directory as a `.py` file.

Question 8:

To solve this question, you should edit the file `irlc/exam/midterm2023a/question_pid.py`. Your task is to implement a discrete PID control in a setting where you are given a **sequence** of states $x_0, \dots, x_{N-1}$, and have to compute the next action u_{N-1} . To avoid boundary issues, you can assume that $N \geq 4$. The discretization time is fixed at $\Delta = 1$.

The functions below all accept a sequence of states `xs`, represented as a list of numbers, and must compute the next action, represented as a `float`.

- (a) Complete `def a_pid_Kp(xs : list, xstar : float, Kp : float)`: Implement PID control to the sequence-of-states setting. The function is given a sequence of N states `xs` and a value of the proportionality term K_p , and should compute the next action. The parameter `xstar` corresponds to a target state x^* , but in this problem we assume that $x^* = K_i = K_d = 0$. The function should return a single real number corresponding to the next action u_{N-1} .
- (b) Complete `def b_pid_full(xs : list, xstar : float, Kp : float, Ki : float, Kd : float)`: This function is similar to the one you implemented in the previous question, but should also take the derivative term K_d , integral term K_i and target x^* into account.
- (c) Complete `def c_pid_stable(xs : list, xstar : float, Kp : float, Ki : float, Kd : float)`: This problem assumes that the states x_k are noisy and that we are particularly concerned about how this affects the derivative term in the PID controller.

Recall the contribution from the derivative term at time step k to the PID controller has the form:

$$K_p \cdot \{\dots\} + K_d e'_k + K_i \cdot \{\dots\}, \quad \text{where} \quad e'_k = \frac{e(t_k) - e(t_k - \Delta)}{\Delta}.$$

I.e., the term e'_k corresponds to the estimated derivative of the error. You should implement a variant of PID control where this term has been replaced with the average estimate over the last two time steps. The contribution from the K_d term should therefore be modified to be:

$$K_d \frac{e'_{k-1} + e'_k}{2}.$$

Solution:

The solution can be found in this directory as a `.py` file.

References

[Her24] Tue Herlau. Sequential decision making. (Freely available online), 2024.

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